

Temporal Fusion Transformer Variants for Residential Energy Consumption Forecasting: An Ablation Study with Wavelet Decomposition and Dimensionality Reduction

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Abstract—Predicting residential electricity consumption at meter level is a challenging time-series problem since household demand is shaped by multiple factors such as long-range weekly rhythms as well as transient events. In this paper we present a comparative study of four Temporal Fusion Transformer model variants for residential energy consumption forecasting. Starting from a baseline TFT model with engineered calendar and power features (V1), we improved the input space using Principle Component Analysis technique i.e. PCA(V2) by which we retained 12 feature components. We also trained a TFT model which made use of both PCA as well as Bayesian Hyperparameter search for hyperparameter tuning(V3). We then proceeded to improve the input space with Discrete Wavelet Transform (DWT) sub band statistics and combined them with PCA in the final model (V4). Under the same full-dataset GPU conditions as V1, V4 surpasses the baseline: test MAE 78.57, RMSE 257.02, R^2 0.8674, MAPE 23.58% versus the baseline’s 79.19, 259.44, 0.8649, and 27.05%. The MAPE improvement of 12.8% is the most pronounced gain. If we compare to the prior LSTM-SVR study conducted on the UCI Steel Industry dataset [1], TFT is better suited to residential forecasting because it utilizes the attention mechanism.

Index Terms—Energy Forecasting, Temporal Fusion Transformer, Wavelet Decomposition, Principle Component Analysis, Residential Metering

I. INTRODUCTION

Predicting electricity consumption at the household level is a major part of making smart grids and renewable energy integration actually work. Unlike industrial loads, which are usually steady, residential data is messy. It’s shaped by human behavior such as daily routines, weekly rhythms, and random spikes from heavy appliances, making it a challenging signal to model.

Traditional methods like ARIMA usually struggle here because residential data is not linear or steady. In a previous part of this project series, SVR and LSTM models were tested on industrial steel consumption data [1], where SVR performed well due to the smaller, smoother dataset. However, SVR does not scale well when you have to handle multiple meters at once, which is the reality for most residential setups.

The Temporal Fusion Transformer (TFT) [2] is designed specifically for this kind of multi-horizon forecasting. It han-

dles different types of inputs like calendar data we know in advance and power measurements we do not and uses an attention mechanism to figure out which features actually matter for the prediction.

To push the performance further, we can look at frequency-domain features using Discrete Wavelet Transform (DWT). Wavelets are great at breaking down a signal into sub-bands, separating long-term trends from sudden peaks. We can also use Principal Component Analysis (PCA) to clean up redundancy in the data and Bayesian optimization (via Optuna) to stop guessing our hyperparameters and start tuning them systematically.

This paper does three things. First, we build a full TFT pipeline for the THC residential dataset. Second, we test four model versions: a baseline (V1), a version with PCA (V2), a hyperparameter-tuned version using Optuna (V3), and a final version combining wavelets and PCA (V4). Finally, we break down the results and compare them to the previous LSTM-SVR study [1] to see where TFT actually adds value.

II. RELATED WORK

A. LSTM and SVR Baselines

The prior study in this project series evaluated LSTM networks and SVR on the UCI Steel Industry Energy Consumption dataset [1]. SVR with RBF kernel and feature scaling substantially outperformed all LSTM variants (SVR $R^2 = 0.9980$, best LSTM $R^2 = 0.5425$). The gap was attributed to the moderate dataset size (35,040 samples) and the smooth, low-dimensional nature of industrial load profiles, which kernel methods capture well without deep sequential modelling.

B. Temporal Fusion Transformers

Lim et al. [2] proposed TFT as a hybrid joining together recurrent layers for local context with self-attention for long term dependencies. Its explicit partitioning of static, known future, and unknown future inputs makes it naturally suited to metered energy data where calendar features are available ahead of forecast time.

C. Wavelet-Enhanced Forecasting

Discrete wavelet decomposition has been combined with neural networks in energy forecasting to provide frequency localised context. The DWT separates a signal into an approximation sub band and multiple detail sub bands at progressively finer scales. Per meter sub band statistics appended as features enrich the model without changing the architecture.

III. DATASET AND PREPROCESSING

A. THC Residential Consumption Dataset

Experiments use a multi meter residential dataset collected at minute level resolution spanning August 2019 to October 2020 from Uruguay, with eleven monthly CSV files concatenated and aligned on a common datetime index. Each row contains per-phase active power (`apower_ph1`), reactive power, phase voltage, current, power factor, and cumulative energy for one meter reading. After concatenation the dataset contains approximately 144,256 rows across multiple meter identifiers. Missing values were imputed meter wise using forward-fill followed by backward-fill. Redundant phase-2 and phase-3 energy columns were dropped.

B. Feature Engineering

A shared base feature set is built identically for all three variants: apparent power ($S = V \cdot I$), power factor angle ($\theta = \arccos(\text{pf})$), energy intensity, reactive-to-active power ratio (Q/P), hour-of-day, day-of-week, binary weekend and peak-hours (08:00–20:00) flags, and a three-step rolling mean of active power.

C. Train / Validation / Test Split

Data is split chronologically per meter to prevent leakage: 70% training, up to 85% validation (cumulative), full series as test. This yields approximately 100,977, 122,615 and 144,256 samples respectively.

IV. METHODOLOGY

Table I lists the four variants and the main difference between each one. The sections below describe each in more detail.

TABLE I
SUMMARY OF THE FOUR TFT VARIANTS

Model	Feature pipeline	Features	Params
V1 Baseline	18 engineered, no transform	18	115.5k
V2 PCA	V1 + 12 PCA components	30	127k
V3 PCA + Optuna	Same as V2, Optuna-tuned arch.	30	89k
V4 Wav + PCA	DWT + PCA (12 PC) + cyclical time	42	178.6k

A. V1: Baseline TFT

V1 takes the 18-feature engineered set as input with no further transformation. The feature set consists of nine raw electrical measurements (active energy total and per phase, reactive energy total and per phase, active power, reactive power, current, power factor, voltage) and five derived quantities: apparent power, power factor angle, energy intensity, reactive-to-active power ratio, and a 3-step rolling mean of active power. Four binary calendar flags round out the set: hour of day, day of week, weekend indicator, and peak-hours indicator (08:00–20:00).

Inside TFT, these are split into known-real inputs (time index plus the four calendar flags) and unknown-real inputs (the nine measurements and five derived features). The architecture uses hidden size 32, four attention heads, dropout 0.1, and hidden-continuous size 16. Output is 7-quantile via `QuantileLoss`. Each meter is normalised separately using `GroupNormalizer`. The encoder window is 24 timesteps (12 hours of 30-minute readings). Training used the Adam optimiser at 10^{-3} , gradient clipping 0.1, early stopping with patience 5, and a cap of 15 epochs. The model has 115.5k trainable parameters. V1 ran on a Kaggle T4×2 instance; the Lightning trainer used one of the two available GPUs.

B. V2: PCA-Enhanced TFT

V2 keeps the same TFT architecture and training setup as V1, adding 12 PCA components computed from the 14 continuous base features (all features except the four binary calendar flags). The scaler and PCA transform are fitted on the training partition only. Validation and test partitions are projected using that same fitted transform, so no future information enters the earlier splits.

The 12 PCA components are appended to the 18 original features, giving 30 time-varying inputs in total. The motivation was that several base features are strongly correlated such as apparent power, active power, and current move together closely and PCA projections should help TFT’s variable selection network by decorrelating the redundant directions. V2 ran on a single local GPU.

C. V3: PCA with Bayesian Hyperparameter Search

V3 starts from the same 30-feature set as V2 and adds Optuna-driven hyperparameter search. The search space is: learning rate in $[10^{-4}, 10^{-2}]$ (log-uniform), hidden size in $\{16, 32, 64\}$, attention head count in $\{1, 4\}$, dropout in $[0.1, 0.3]$, and hidden-continuous size in $\{8, 16, 32\}$. Each trial ran up to 15 epochs with early stopping; Optuna minimised the best validation loss seen during the trial.

The study selected: learning rate 8.03×10^{-3} , hidden size 16, one attention head, dropout 0.195, hidden-continuous size 16. That gives roughly 89k parameters, well below V1’s 115.5k. The consequences of this choice are discussed in Section V. V3 ran on the same local GPU as V2.

D. V4: Wavelet plus PCA TFT

V4 combines wavelet features and PCA under the same training conditions as V1. It ran on a Kaggle T4×2 instance (one GPU used) on the full 144,256-row dataset across all 14 meters, with the same 15-epoch budget.

After extracting DWT sub-band statistics, PCA is fitted on all continuous wavelet-augmented features using only the training partition. Twelve principal components are retained, explaining 97.59% of the combined feature variance. Two cyclical time encodings are added to the known-real set:

$$h_{\sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad h_{\cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right) \quad (1)$$

Raw integer hour (0–23) creates a discontinuity between hour 23 and hour 0 in feature space. The sine/cosine pair removes this by wrapping the clock around a circle. The total feature count reaches 42, and the model grows to 178.6k parameters because of the wider input projection. Everything else – optimiser, learning rate, gradient clipping, encoder length, early stopping, normalisation – is identical to V1.

E. Evaluation

All variants are evaluated on the same held-out validation and test partitions using four metrics:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad \text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i + \epsilon} \right| \quad (3)$$

MAPE deserves particular attention for residential load. Household consumption swings from near-zero overnight values to afternoon peaks, so a model that makes proportionally large errors on low-consumption periods will score badly on MAPE even if its absolute errors are modest. That makes MAPE a more honest measure of practical accuracy than MAE or RMSE alone for this dataset.

V. RESULTS AND DISCUSSION

A. Full Metric Table

Table II reports validation and test metrics for all four variants.

B. Notebook output visual comparison

The figures below were extracted directly from executed notebook outputs in the corresponding TFT Notebooks subdirectories. They are included as run evidence rather than redrawn illustrations.

C. Feature Pipeline Overview

Fig. 3 shows how each variant derives its feature set from the raw data.

TABLE II
PERFORMANCE OF ALL FOUR TFT VARIANTS (VALIDATION AND TEST)

Model	MAE	RMSE	R ²	MAPE (%)
<i>Validation set</i>				
V1 Baseline	82.01	264.25	0.8680	26.44
V2 PCA	83.20	261.64	0.8706	28.25
V3 PCA + Optuna	86.40	265.01	0.8673	26.42
V4 Wav+PCA	81.73	261.68	0.8706	23.67
<i>Test set</i>				
V1 Baseline	79.19	259.44	0.8649	27.05
V2 PCA	80.33	257.00	0.8675	28.93
V3 PCA + Optuna	84.43	261.28	0.8630	27.76
V4 Wav+PCA	78.57	257.02	0.8674	23.58

D. V1 vs. V2: PCA Without Wavelets Gives Mixed Results

V2 and V1 are close on aggregate error. V2 slightly improves RMSE and R² (257.00 vs. 259.44, and 0.8675 vs. 0.8649), but MAE and MAPE get worse (80.33 vs. 79.19, and 28.93% vs. 27.05%). So PCA alone is not a clear net win on this dataset.

The likely reason is that this PCA step is close to a rotation, not a strong compression. We keep 12 components out of 14 continuous inputs, then append those components to the original feature set. That expands the input space while keeping substantial redundancy. The traces in Fig. 1 are therefore very similar to V1.

E. V1 vs. V3: What Happens When Optuna Picks Too Small a Model

In this run, Optuna did almost nothing useful for predictive quality. V3 is slightly worse than V1 on all four metrics, and it also fails to beat V2 on MAE, RMSE, and R². The long-window traces for V2 and V3 in Fig. 1 look nearly identical.

The selected setup (hidden size 16, one attention head, 89k parameters) likely underfit the full 14-meter dataset. This is a common failure mode with early stopping: smaller models can look good in early epochs and then plateau.

The practical fix is to keep Bayesian search but enforce a capacity floor, for example hidden size ≥ 32 and more than one attention head.

F. V1 vs. V4: A clean comparison

V4 is directly comparable to V1 because both use the same full dataset, meter count, training budget, and GPU setup. V4 improves all four test metrics over V1:

- MAE: 79.19 $\rightarrow$ 78.57 (a 0.8% drop)
- RMSE: 259.44 $\rightarrow$ 257.02 (a 0.9% drop)
- R²: 0.8649 $\rightarrow$ 0.8674 (+0.25 percentage points)
- MAPE: 27.05% $\rightarrow$ 23.58% (a 12.8% relative reduction)

The gain is not uniform across metrics. RMSE and R² improve only slightly, while MAPE drops much more. So the final model helps most on proportional error, especially during lower-load periods.

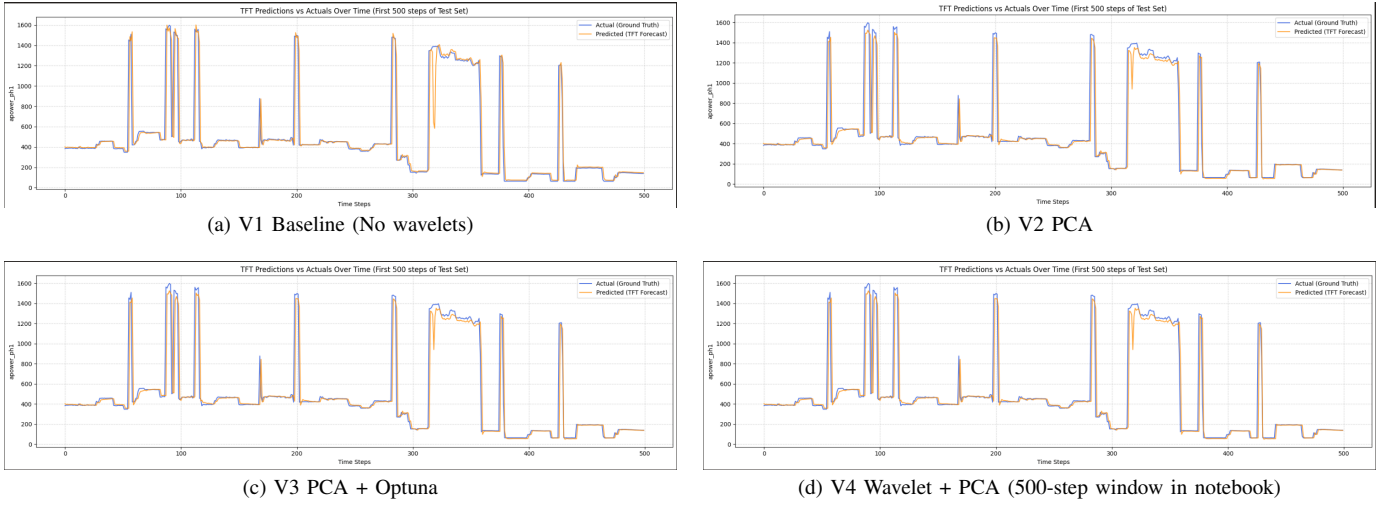


Fig. 1. Prediction traces extracted from notebook outputs. V1, V2, and V3 are nearly overlapping on long windows; V4 also tracks similarly on its stored 500-step view.

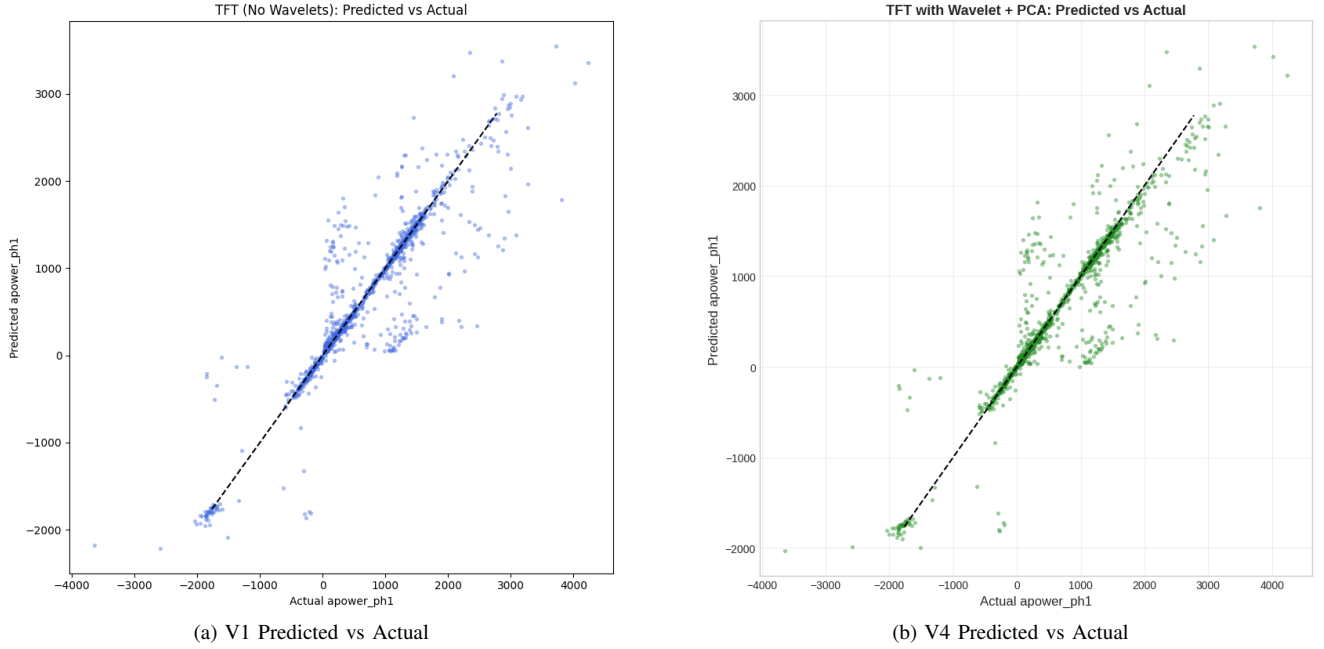


Fig. 2. Scatter plots from notebook outputs for the baseline and the final wavelet+PCA run. Both align closely with the diagonal reference.

G. V2 vs. V4: Did wavelets help enough?

This is the most direct wavelet test, since both variants already include PCA. On test data, V2 and V4 are effectively tied on RMSE and R^2 :

- RMSE: 257.00 (V2) vs. 257.02 (V4)
- R^2 : 0.8675 (V2) vs. 0.8674 (V4)

So if RMSE or R^2 is the primary objective, wavelet features add little over PCA in this experiment. The clear difference is MAPE (28.93% in V2 vs. 23.58% in V4), where V4 is substantially better.

This split result matches the notebook visuals in Fig. 1: global shapes are very similar, while improvements appear mostly in smaller local deviations.

H. Comparison with Prior Work

The prior NITK study [1] applied stacking ensembles (LSTM, Random Forest, DNN, Evolutionary Trees, combined by XGBoost) to a single industrial series. Their best ensemble reached $R^2 = 0.97$ – 0.98 and RMSE around 636. Our V4 reaches $R^2 = 0.8674$ on a substantially harder target.

The difference is in the data, not the model. Steel plant load follows a production schedule: it is smooth, predictable, and driven by process constraints. Residential load is driven by individual household decisions. Two meters in the same building can have very different profiles depending on occupant behaviour. The MAPE range of 23–27% in our results is not poor modelling, it reflects genuine unpredictability in the signal.

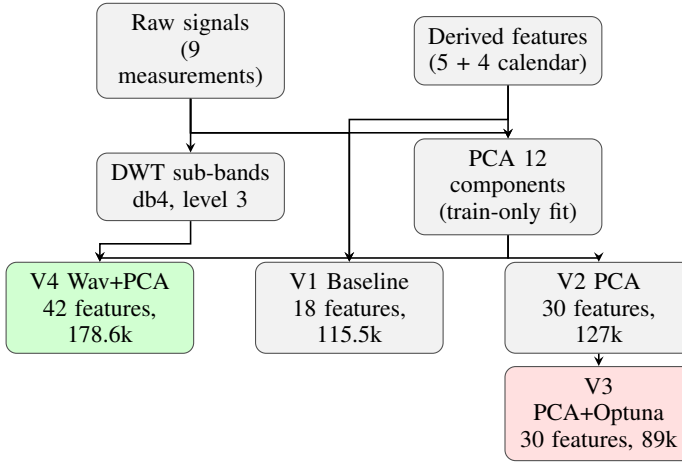


Fig. 3. How each variant builds its feature set. Green = best performing (V4). Red = underperforms baseline (V3).

Structurally, TFT handles 14 meters in one pass through static categorical embeddings. SVR would need one model per meter. TFT’s attention weights also give a per-feature and per-timestep importance readout that SVR cannot provide. For multi-meter residential data these are practical advantages, not just theoretical ones.

I. MAPE Comparison Across Variants

Fig. 4 shows the test-set MAPE for all four variants. V4 is the only one to dip below 24%.

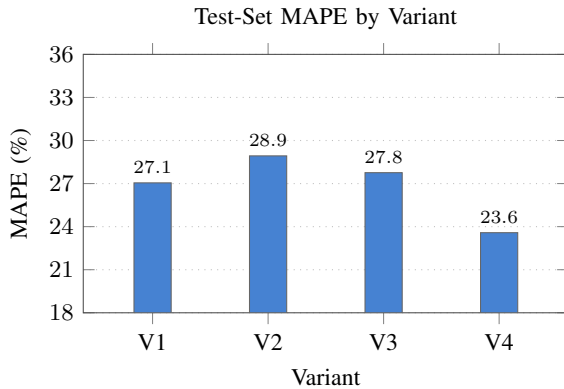


Fig. 4. Test-set MAPE for all variants. V4 achieves the lowest MAPE at 23.58%.

VI. WHAT MAKES THIS APPROACH DIFFERENT

Several design choices separate this work from the prior NITK study and from common energy forecasting practice.

One model, many meters. TFT trains a single model across all 14 residential meters using static meter-ID embeddings. This avoids the combinatorial cost of training and maintaining one model per entity which is a meaningful practical difference when the number of meters scales to hundreds.

Wavelet statistics as an optional frequency view. DWT sub-band features add frequency-domain context without

changing TFT architecture. In our runs, this did not materially change RMSE/ R^2 over PCA-only inputs, but it helped MAPE when combined with PCA under matched conditions.

Leakage-free PCA. The scaler and PCA are fitted on the training partition only. Some published studies fit PCA on the full dataset before splitting, which leaks validation and test distribution statistics into the training pipeline. Our implementation avoids this.

Cyclical time encodings. Hour of day as an integer from 0 to 23 creates a spurious distance of 23 between the last hour of one day and the first hour of the next. Sine and cosine encodings wrap this correctly. The improvement is small but costs nothing.

VII. CONCLUSION

Four TFT variants were tested on the THC multi-meter residential dataset (144,256 rows, 14 meters, 30-minute intervals). The main findings are summarised below.

V1 established the baseline at test $R^2 = 0.8649$ and MAPE = 27.05%. V2 (PCA) gave mixed results: slightly better RMSE/ R^2 , but worse MAE/MAPE. V3 (PCA + Optuna) reduced model size but did not improve predictive quality and was generally close to, or worse than, V2 and V1. V4 (wavelet + PCA), trained under the same conditions as V1, improved all baseline metrics, with the strongest gain in MAPE.

Two practical takeaways matter most. First, wavelet features were not a universal win: they had little effect on RMSE/ R^2 compared with PCA-only inputs, but they helped percentage error. Second, Optuna search as configured here was not useful; capacity constraints in the search space need tightening before drawing conclusions from HPO.

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REFERENCES

- [1] S. Reddy, S. Akashdeep, R. Harshvardhan, and S. S. Kamath, “Stacking deep learning and machine learning models for short-term energy consumption forecasting,” *Advanced Engineering Informatics*, vol. 52, p. 101542, 2022.
- [2] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, “Temporal fusion transformers for interpretable multi-horizon time series forecasting,” *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
- [3] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [4] H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, and V. Vapnik, “Support vector regression machines,” in *Advances in Neural Information Processing Systems*, vol. 9, 1997.
- [5] S. G. Mallat, “A theory for multiresolution signal decomposition: the wavelet representation,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 7, pp. 674–693, 1989.
- [6] M. S. Al-Musaylh, R. C. Deo, J. F. Adamowski, and Y. Li, “Short-term electricity demand forecasting with MARS, SVR and ARIMA models using aggregated demand data in Queensland, Australia,” *Advanced Engineering Informatics*, vol. 35, pp. 1–16, 2018.

- [7] J. Moon, S. Jung, J. Rew, S. Rho, and E. Hwang, "Combination of short-term load forecasting models based on a stacking ensemble approach," *Energy and Buildings*, vol. 216, p. 109921, 2020.
- [8] S. Santoso, E. J. Powers, W. M. Grady, and P. Hofmann, "Power quality assessment via wavelet transform analysis," *IEEE Trans. Power Del.*, vol. 11, no. 2, pp. 924–930, 1996.